

The screenshot shows the VS Code interface with the Explorer on the left, displaying the project structure for 'BookFlow_Exp_6'. The main editor area shows the terminal output of the following commands:

```
(venv) PS D:\DBSE-Practical\BookFlow_Exp_6> pip install fastapi uvicorn
Downloading anyio-4.15.1-py3-none-any.whl (132 kB)
132.1/132.1 KB 433.6 KB/s eta 0:00:00
Collecting exceptiongroup==1.0.2
Downloading exceptiongroup-1.3.1-py3-none-any.whl (16 kB)
Collecting idna>=2.8
Downloading idna-3.20-py3-none-any.whl (69 kB)
69.6/69.6 KB 474.7 KB/s eta 0:00:00
Installing collected packages: typing-extensions, idna, h11, click, annotated-types, annotated-doc, uvicorn, typing-inspection, pydantic-core, exceptiongroup, pydantic, anyio, starlette, fastapi
Successfully installed annotated-doc-0.0.5 annotated-types-0.8.0 anyio-4.15.1 click-8.5.0 exceptiongroup-1.3.1 fastapi-0.141.1 h11-0.16.0 idna-3.20 pydantic-2.13.5 pydantic-core-2.46.5 starlette-1.7.0 typing-extensions-4.16.0 typing-inspection-0.4.4 uvicorn-0.54.0
WARNING: You are using pip version 22.0.4; however, version 26.2.1 is available.
You should consider upgrading via the 'D:\DBSE-Practical\BookFlow_Exp_6\venv\Scripts\python.exe -m pip install --upgrade pip' command.
(venv) PS D:\DBSE-Practical\BookFlow_Exp_6> pip show fastapi
Name: fastapi
Version: 0.141.1
Summary: FastAPI framework, high performance, easy to learn, fast to code, ready for production
Home-page:
Author:
Author-email: =?utf-8?q?Sebasti=C3=A1n_Ram=C3=A0rez?= <tiangolc@gmail.com>
License:
Location: d:\dbse-practical\bookflow_exp_6\venv\lib\site-packages
Requires: annotated-doc, pydantic, starlette, typing-extensions, typing-inspection
Required-by:
(venv) PS D:\DBSE-Practical\BookFlow_Exp_6> pip show uvicorn
Name: uvicorn
Version: 0.54.0
Summary: The lightning-fast ASGI server.
Home-page:
Author:
Author-email: Tom Christie <tom@tomchristie.com>
License:
Location: d:\dbse-practical\bookflow_exp_6\venv\lib\site-packages
Requires: click, h11, typing-extensions
Required-by:
```

The screenshot shows the VS Code interface with the Explorer on the left, displaying the project structure for 'BookFlow_Exp_6'. The main editor area shows the code in 'main.py' and the terminal output of the uvicorn server.

main.py

```
1 from fastapi import FastAPI
2 from routes.catalog import router as catalog_router
3 from routes.admin import router as admin_router
4
5 app = FastAPI(
6     title="BookFlow Library API"
7 )
8
9 app.include_router(catalog_router)
10 app.include_router(admin_router)
```

Terminal Output

```
(venv) PS D:\DBSE-Practical\BookFlow_Exp_6> New-Item routes\admin.py
-----
-a----      26-09-2026      08:58                0 admin.py
-----

(venv) PS D:\DBSE-Practical\BookFlow_Exp_6> uvicorn main:app --reload
INFO: Will watch for changes in these directories: ['D:\DBSE-Practical\BookFlow_Exp_6']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [25420] using StatReload
INFO: Started server process [24988]
INFO: Waiting for application startup.
INFO: Application startup complete.
```

BookFlow Library API

0.1.0

OAS 3.1

/openapi.json

default

GET

/books

Get Books

GET

/books/{book_id}

Get Book

POST

/admin/books

Add Book

PUT

/admin/books/{book_id}

Update Book

DELETE

/admin/books/{book_id}

Delete Book

Schemas

Book

>

Expand all

object

HTTPValidationError

>

Expand all

object

ValidationError

>

Expand all

object

default

GET

/books

Get Books

Parameters

No parameters

Execute

Clear

Cancel

Responses

Curl

curl -X 'GET' \

'http://127.0.0.1:8000/books' \

-H 'accept: */*'

Request URL

http://127.0.0.1:8000/books

Server response

Code

Details

200

Response body

{

"id": 101,

"title": "Python Programming",

"author": "John Smith",

"price": 450,

"isbn": "9781234567890",

"status": "Available"

},

"id": 102,

"title": "Database Management",

"author": "Robert Brown",

"price": 350,

"isbn": "9789876543210",

"status": "Available"

}

Response headers

content-length: 240

content-type: application/json

date: Sat, 6 Sep 2020 03:34:10 GMT

server: uvicorn

Download

GET

/books

Get Books

GET

/books/{book_id}

Get Book

Parameters

Cancel

Name	Description
book_id required	
integer	
(path)	101

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/books/101' \
  -H 'accept: */*'
```

Request URL

http://127.0.0.1:8000/books/101

Server response

Code	Details
200	Response body <pre>{ "id": 101, "title": "Python Programming", "author": "John Smith", "price": 500, "isbn": "9781234567890", "status": "Available" }</pre> Download Response headers <pre>content-length: 117 content-type: application/json date: Sat, 26 Sep 2026 03:17:17 GMT server: uvicorn</pre>

Responses

GET

/books

Get Books

GET

/books/{book_id}

Get Book

Parameters

Name	Description
book_id required	
integer	
(path)	999

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/books/999' \
  -H 'accept: */*'
```

Request URL

http://127.0.0.1:8000/books/999

Server response

Code	Details
404	Error: Not Found Response body <pre>{ "detail": "Book not found" }</pre> Response headers <pre>content-length: 27 content-type: application/json date: Sat, 26 Sep 2026 03:38:14 GMT server: uvicorn</pre>

Responses

Code	Description
------	-------------

POST

/admin/books Add Book

Cancel

Parameters

No parameters

Request body required application

Edit Value | Schema

```
{  "title": "FastAPI for Beginners",  "author": "A. Kumar",  "price": 650,  "isbn": "978112223334",  "status": "Available"}}
```

Execute Clear

Responses

Curl

```
curl -X 'POST' \  'http://127.0.0.1:8000/admin/books' \  -H 'accept: */*' \  -H 'content-type: application/json' \  -d '{  "title": "FastAPI for Beginners",  "author": "A. Kumar",  "price": 650,  "isbn": "978112223334",  "status": "Available"}'
```

Request URL

http://127.0.0.1:8000/admin/books

Server response

Code Details

200

Response body

```
{  "message": "Book added successfully",  "book": {    "id": 101,    "title": "FastAPI for Beginners",    "author": "A. Kumar",    "price": 650,    "isbn": "978112223334",    "status": "Available"  }}
```

GET

/books Get Books

Parameters

No parameters

Execute

Responses

Curl

```
curl -X 'GET' \  'http://127.0.0.1:8000/books' \  -H 'accept: */*'
```

Request URL

http://127.0.0.1:8000/books

Server response

Code Details

200

Response body

```
{  {    "id": 101,    "title": "python Programming",    "author": "John Smith",    "price": 450,    "isbn": "9781234567890",    "status": "Available"  },  {    "id": 102,    "title": "Database Management",    "author": "Robert Brown",    "price": 550,    "isbn": "9789876543210",    "status": "Available"  },  {    "id": 103,    "title": "FastAPI for Beginners",    "author": "A. Kumar",    "price": 650,    "isbn": "978112223334",    "status": "Available"  }}
```

Response headers

```
Curl -X "PUT" \
  https://127.0.0.1:8000/admin/books/103/ \
  -H "accept: */*" \
  -H "Content-Type: application/json" \
  -d '{
    "title": "FastAPI For Beginners - 2nd Edition",
    "author": "A. Kumar",
    "price": 700,
    "isbn": "9781112223314",
    "status": "Available"
  }'
```

Request URL

http://127.0.0.1:8000/admin/books/103

Server response

Code	Details
200	Response body <pre>{ "message": "Book updated successfully", "book": { "id": 103, "title": "FastAPI For Beginners - 2nd Edition", "author": "A. Kumar", "price": 700, "isbn": "9781112223314", "status": "Available" } }</pre>

GET

/books/{book_id} Get Book

Cancel

Parameters

Name	Description
book_id * required integer (optional)	103

Execute

Clear

Responses

curl -s 'http://127.0.0.1:8000/books/103' \

-H 'accept: */*' \

Request URL

http://127.0.0.1:8000/books/103

Server response

Code

Details

200

Response body

```
{
  "id": 103,
  "title": "Python for Beginners - 2nd Edition",
  "author": "A. Kumar",
  "price": 799,
  "isbn": "9781122333334",
  "status": "Available"
}
```

Download

Response headers

```
content-length: 134
content-type: application/json
date: Sat, 26 Sep 2026 04:03:41 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	No links

Media type

application/json

Controls Accept header

Example Value | Schema

DELETE

/admin/books/{book_id} Delete Book

Parameters

Name	Description
book_id * required	
integer	103
(path)	

ExecuteClear

Responses

Curl

curl -X 'DELETE' \n'http://127.0.0.1:8000/admin/books/103' \n-H 'accept: */*'

Request URL

http://127.0.0.1:8000/admin/books/103

Server response

Code

Details

200

Response body

{\n "message": "Book deleted successfully",\n "book": {\n "id": 103,\n "title": "FastAPI for Beginners - 2nd Edition",\n "author": "A. Kumar",\n "price": 700,\n "isbn": "978112223334",\n "status": "Available"\n }\n}

Response headers

content-length: 181\ncontent-type: application/json\ndate: Sat, 26 Sep 2026 04:04:59 GMT\nserver: uvicorn

Responses

Code

Description

200

Successful Response

Media type

application/json

Controls Accept header.

Example Value | Schema

"string"

422

Validation Error

Media type

application/json

Example Value | Schema

{\n "detail": {\n {\n "loc": [\n "string",\n 0\n],\n "msg": "string",\n "type": "string",\n "input": "string",\n "ctx": {}\n }\n }\n}

GET

/books Get Books

Parameters

No parameters

ExecuteClear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/books' \
  -H 'accept: */*'

```

Request URL

```
http://127.0.0.1:8000/books

```

Server response

Code

Details

200

Response body

```
{
  {
    "id": 101,
    "title": "Python Programming",
    "author": "John Smith",
    "price": 450,
    "isbn": "9781234567890",
    "status": "Available"
  },
  {
    "id": 102,
    "title": "Database Management",
    "author": "Robert Brown",
    "price": 550,
    "isbn": "9789876543210",
    "status": "Available"
  }
}

```

Response headers

```
content-length: 240
content-type: application/json
date: Sat, 26 Sep 2026 04:06:59 GMT
server: uvicorn

```

Responses

Code

Description

200

Successful Response

Media type

application/json

Controls Accept header

POST

/admin/books Add Book

Parameters

No parameters

CancelReset

Request body ^{required}

application/json

Edit Value | Schema

```
{
  "title": "Invalid Book",
  "author": "Test Author",
  "price": -100,
  "isbn": "9781234567890",
  "status": "Available"
}
```

ExecuteClear

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/admin/books' \
  -H 'accept: */*' \
  -H 'content-type: application/json' \
  -d '{
    "title": "Invalid Book",
    "author": "Test Author",
    "price": -100,
    "isbn": "9781234567890",
    "status": "Available"
  }'

```

Request URL

```
http://127.0.0.1:8000/admin/books

```

Server response

Code

Details

422

Error: Unprocessable Entity

Response body

```
{
  "detail": [
    {
      "type": "greater_than_equal",
      "loc": [
        "price"
      ]
    }
  ]
}

```

Result: The BookFlow Library REST API was successfully implemented using FastAPI and APIRouter. GET, POST, PUT, and DELETE operations were tested successfully through Swagger UI. Pydantic validation was also verified using an invalid negative price, which produced a 422 validation error. The application uses a shared in-memory book collection for storing data temporarily